
id: MDCS-TICKET-004
title: Create Oracle SQL reference drawers.
epic: epic-dumb-copilot
status: ready
risk: low
allowed_areas:
- references/oracle-sql/
must_not_touch:
- tools/
- procedures/
requirements:
- Create the Oracle SQL reference directory structure: `references/oracle-sql/sections/`.
- Create `references/oracle-sql/README.md` as the index. It must:
* Declares the reference scope and consuming skills.
* Limit the index size to at most 120 lines total.
* Contain a markdown Sections table with columns: `id` | `topic` | `open when` (note the exact space in “open when”).
* Map the following drawers: `query_tuning`, `plsql_basics`, and `oracle_idioms`.
- Create `references/oracle-sql/sections/query_tuning.md`:
* Limit size to at most 150 lines total.
* Begin with a one-line **relevant-when**: header key-style line.
* Focus on Oracle explain plan analysis (interpreting Table Access Full, Index Range Scan, Nested Loops, Hash Joins).
* Highlight Oracle-specific tuning hints (`/*+ INDEX */`, `/*+ LEADING */`, `/*+ PARALLEL */`).
* Explain subquery factoring (CTEs) vs. subqueries in Oracle.
- Create `references/oracle-sql/sections/plsql_basics.md`:
* Limit size to at most 150 lines total.
* Begin with a one-line **relevant-when**: header key-style line.
* Explain cursor handling, exceptions, and bulk operations (`BULK COLLECT`, `FORALL`).
* Demonstrate autonomous transactions (`PRAGMA AUTONOMOUS_TRANSACTION`).
- Create `references/oracle-sql/sections/oracle_idioms.md`:
* Limit size to at most 150 lines total.
* Begin with a one-line **relevant-when**: header key-style line.
* Detail analytical window functions (`ROW_NUMBER`, `RANK`, `DENSE_RANK`, `LEAD`, `LAG`, `FIRST_VALUE`, `LISTAGG`).
* Cover Oracle date arithmetic (e.g., handling `SYSDATE`, intervals, and timezone offsets).
* Demonstrate hierarchical queries using `START WITH ... CONNECT BY PRIOR` and recursive CTEs.
non_goals:
- Do not include project-private schema names, login details, or specific table structures. Keep references strictly generic and dia
acceptance_criteria:
- Running `python tools/validate_mdcs_library.py` passes on the Oracle SQL references.
- The index is 120 lines and sections are 150 lines.
- Every drawer begins with a one-line **relevant-when**: header.
verification_commands:
- `python tools/validate_mdcs_library.py`
depends_on:
- MDCS-TICKET-001

Body

Goal

Compile standard, dense Oracle SQL reference drawers to feed Copilot during active procedures, preventing dialect confusion.

Context

Web-based Copilots often suggest generic SQL Server or PostgreSQL syntax (e.g. `LIMIT` instead of `ROWNUM/FETCH FIRST`, or generic interval math) which fails in Oracle SQL. Dedicated drawers teach the model Oracle-specific behavior.

Procedure

1. Initialize `references/oracle-sql/README.md` and write the index table.
2. Draft `references/oracle-sql/sections/query_tuning.md` starting with a `relevant-when:` header and focusing on explaining Oracle's optimizer choices and hints.
3. Draft `references/oracle-sql/sections/plsql_basics.md` starting with a `relevant-when:` header and highlighting bulk binding and cursors.
4. Draft `references/oracle-sql/sections/oracle_idioms.md` starting with a `relevant-when:` header and outlining analytical functions, hierarchical queries, and dates.
5. Run the validator tool to check formatting and file line constraints.